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Measuring apparatus, measuring method, and storage medium

Abstract

A measuring apparatus is configured to acquire information on a target. The measuring apparatus includes an irradiating unit configured to irradiate a specific area of the target with irradiation light, a detecting unit configured to receive exit light from the target which is caused by irradiating the specific area with the irradiation light, and a processing unit configured to process a signal output from the detecting unit. The processing unit causes the irradiating unit to shape a wavefront of the irradiation light by feeding back a first signal output from the detecting unit that has received the exit light. The processing unit acquires information about the specific area using a second signal output from the detecting unit that has received exit light from the target which is by irradiating the specific area with irradiation light having a shaped wavefront.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of International Patent Application No. PCT/JP2021/027930, filed on Jul. 28, 2021, which claims the benefit of Japanese Patent Application No. 2020-173529, filed on Oct. 14, 2020, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

Technical Field

(1) One of the aspects of the disclosure relates to a measuring apparatus, a measuring method, and a storage medium for acquiring information on an object to be measured (referred to as a target hereinafter) using light.

Description of Related Art

(2) Japanese Domestic PCT Publication No. 2014-500751 discloses a method and apparatus for extracting a parameter in blood, such as blood hemoglobin concentration and oxygen saturation, by measuring dynamic changes in blood in a living body using dynamic light scattering (DLS). Japanese Patent No. 6241853 discloses a measuring apparatus for noninvasively measuring a scattering coefficient in an area including a vascular region and calculates the serum lipid concentration from the scattering coefficient.

(3) The methods disclosed in Japanese Domestic Publication No. 2014-500751 and Japanese Patent No. 6241853 can noninvasively and easily measure a target. However, these methods do not shape a wavefront of light during measurement, and thus have difficulty in accurately acquiring information about the target, such as minute changes in physiological parameters in a living body, due to the influence of light scattering.

SUMMARY

(4) One of the aspects of the present disclosure provides a measuring apparatus, a measuring method, and a storage medium, each of which can acquire information on a target with high accuracy using light.

(5) A measuring apparatus according to one aspect of the disclosure is configured to acquire information on a target. The measuring apparatus includes an irradiating unit configured to irradiate a specific area of the target with irradiation light, a detecting unit configured to receive exit light from the target which is caused by irradiating the specific area with the irradiation light, and a processing unit configured to process a signal output from the detecting unit. The processing unit causes the irradiating unit to shape a wavefront of the irradiation light by feeding back a first signal output from the detecting unit that has received the exit light. The processing unit acquires information about the specific area using a second signal output from the detecting unit that has received exit light from the target which is by irradiating the specific area with irradiation light having a shaped wavefront. A measuring method corresponding to the above measuring apparatus and a non-transitory computer-readable storage medium storing a program that causes a computer to execute the measuring method also constitute another aspect of the disclosure.

(6) Further features of the disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of a measuring apparatus according to a first embodiment.
- (2) FIG. 2 is a schematic diagram of a light propagation path in the first embodiment.
- (3) FIG. 3 is a flowchart of a measuring method according to the first and second embodiments.
- (4) FIG. 4 is a schematic diagram of a wavefront shaping unit and a modulation pattern in the first embodiment.
- (5) FIG. 5 illustrates an example of a measurement signal in the first embodiment.
- (6) FIG. 6 explains a feedback signal in the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

(7) Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure.

First Embodiment

- (8) Referring now to FIG. 1, a description will be given of a measuring apparatus according to a first embodiment of the disclosure. FIG. 1 is a block diagram of a measuring apparatus (target measuring apparatus) **100** according to this embodiment. The measuring apparatus **100** according to this embodiment can acquire information on a target and includes a light source **101**, an irradiating unit **110**, a detecting unit **120**, and a processing unit **130**. The irradiating unit **110** irradiates a specific area of a target **190** with light emitted from the light source **101** as irradiation light **151**. The detecting unit **120** receives exit lights **161** and **162** from the target **190** which are caused by irradiating the specific area of the target **190** with the irradiation light **151**. The processing unit **130** processes a signal output from the detecting unit **120** and includes a control unit **131**, a data acquiring unit **132**, a signal processing unit **133**, and a memory **134**.
- (9) The light source **101** includes, for example, a laser that emits light with a wavelength of 300 nm to 1200 nm, which is continuous light (CW light: continuous wave light) whose intensity is constant over time. For example, the wavelength can be selected according to the absorption spectrum of the item to be measured from water, fat, protein, oxygenated hemoglobin, reduced hemoglobin, etc., which are the main constituents of the target **190**. Alternatively, in measuring a scattering coefficient, a wavelength other than the above absorption wavelength may be selected. Depending on the purpose, the wavelength may be selected outside the above range, and the laser may emit intensity-modulated light with an arbitrary frequency or pulsed light. The irradiation light may have a sufficiently long coherence length (such as several centimeters or longer). The intensity of the light irradiated to the target **190** is adjusted within the intensity that can be irradiated under the safe standard.
- (10) Light **150** emitted from the light source **101** is collimated and enters a wavefront shaping unit **111** in the irradiating unit **110**. The wavefront shaping unit **111** includes a spatial light modulator (SLM). The SLM may be a phase modulation type or an amplitude modulation type. The wavefront shaping unit **111** may also be either a transmission type or reflection type SLM. More specifically, the SLM can use an LCOS (Liquid Crystal On Silicon), a DMD (Digital Mirror Device), a GLV (Grating Light Valve), or the like. The SLM may be able to modulate light as fast as possible (such as at several kHz or higher).
- (11) The wavefront shaping unit **111** performs wavefront shaping (WFS), which will be described below, and irradiates the target **190** with light having a shaped wavefront as irradiation light **151**. The wavefront shaping unit **111** shapes the wavefront of the irradiation light **151** by performing phase modulation or amplitude modulation based on a base pattern. At this time, the irradiating unit **110** may include an optical system for irradiating collimated light or converged light of any beam size. Alternatively, the irradiating unit **110** may include an optical fiber, and irradiate the target **190** with the light shaped by the wavefront shaping unit **111** via the optical fiber.
- (12) The exit lights **161** and **162** that have propagated inside the target **190** and exited at different distances from an irradiating position on the surface of the target **190** are received by a detector

(first detector) **121** and a detector (second detector) **122** in the detecting unit **120**, respectively. The detecting unit **120** may include an optical system for condensing the exit lights **161** and **162** from the target **190** onto the detectors **121** and **122**. Alternatively, the exit lights **161** and **162** may be guided to detectors **121** and **122**, respectively, via optical fibers. The detectors **121** and **122** can use sensors such as photodetectors (PD), avalanche photodiodes (APD), and photomultiplier tubes (PMT). Alternatively, a two-dimensional array sensor such as a CMOS or a CCD can be used. The detectors **121** and **122** may respond at high speed (>1 MHz).

(13) The light intensities received by the detectors **121** and **122** are converted into electrical signals, and passed to the data acquiring unit **132**. The data acquiring unit **132** amplifies the signal with an amplifier circuit as necessary, performs analog-to-digital (A/D) conversion for the amplified signal, and generates a digital signal. The signal processing unit **133** performs various signal processing for the digital signal as necessary. The memory **134** properly stores measurement results, intermediate information during measurement, and the like. The memory **134** also stores a measurement flow, measurement conditions, parameters necessary for analysis, and the like. The signal processing unit **133** properly accesses the memory **134** according to the processing, refers to necessary information, and executes the processing.

(14) The control unit **131** controls each module in the light source **101**, the irradiating unit **110**, the detecting unit **120**, and the processing unit (data acquiring unit **132**, signal processing unit **133**, and memory **134**) in the measuring apparatus **100** based on the measurement flow (measuring method), which will be described below.

(15) Referring now to FIG. 2, a description will be given of propagation paths of the irradiation light **151** that is emitted from the irradiating unit **110**, propagates inside the target **190**, and exits. FIG. 2 schematically illustrates the propagation paths of the irradiation light **151** that is emitted from the irradiating unit **110**, propagates inside the target **190**, and exits. The irradiation light **151** is influenced by a scatterer **191** derived from cells and tissue structures inside the target **190** and propagates inside the target **190** while being repeatedly scattered. From the light irradiating position at which the irradiation light **151** enters the target **190**, the path along which the exit light propagates inside the target **190** differs depending on a distance ρ in the horizontal direction. For example, the spreading banana-shaped distribution illustrated in the propagation path **155** is shown at the exit position of the exit light **161**. Similarly, the distribution like the propagation path **156** is shown at the exit position of the exit light **162**.

(16) In a case where the two propagation paths **155** and **156** are compared, the propagation path **156** penetrates and propagates to a deeper area in the target **190**. That is, as the distance ρ between the irradiating position and the detection position increases, the detected exit light is light that has propagated to a deeper position in the target **190** and reflects the optical characteristics of a deeper position inside the target **190**. This propagation path can be estimated based on the absorption coefficient and scattering coefficient, which are the optical characteristics of the target **190**, using the radiation transport equation or the diffusion equation as its diffusion approximation. That is, adjusting the distance ρ can specify the light propagating area in the depth direction inside the target **190**.

(17) The detecting unit **120** can arbitrarily set the detection positions (distance ρ) at which the detectors **121** and **122** receive the lights so as to measure the exit lights from the arbitrary exit positions on the surface of the target **190**. As illustrated in FIG. 2, the distance ρ is set such that the detector **121** detects the exit light **161** that has propagated through a specific area (first area: area that contains no blood vessel (non-vascular area)) near the surface of the target **190**. On the other hand, the distance ρ is set such that the detector **122** detects the exit light **162** that has propagated through a vascular region **192** (second area: area that contains a blood vessel (vascular area)) which is a relatively deep specific area in the target **190**.

(18) Referring now to FIG. 3, a description will be given of the measuring method according to this embodiment. FIG. 3 is a flowchart of the measuring method according to this embodiment. Each

step in FIG. 3 is mainly executed by the irradiating unit **110**, the detecting unit **120**, and the processing unit **130** in the measuring apparatus **100**.

(19) First, in step **S10**, the processing unit **130** measures the exit light **161** and performs WFS based on the signal intensity of the light that has propagated through the first area (non-vascular area). The WFS shapes (optimizes) the wavefront (phase distribution or amplitude distribution) of the irradiation light **151** incident on the target **190** so as to maximize or minimize the monitored feedback signal (first signal). More specifically, step **S10** includes steps **S11** and **S12**.

(20) First, in step **S11**, the processing unit **130** measures the optical intensity measured by the detector **121** as a feedback signal. This light intensity is the average value of the light intensities received by the detector **121** for a certain period. Next, in step **S12**, the processing unit **130** performs wavefront shaping for the irradiation light **151** using the wavefront shaping unit **111**. That is, the processing unit **130** feeds back the first signal (first feedback signal) from the detecting unit **120** that has received the exit light **161** that has propagated through a non-vascular area (first area) and exited from the target **190**, and causes the irradiating unit **110** to shape the wavefront of the irradiation light **151**. The wavefront shaping can be performed based on a modulation pattern (base pattern), such as a Hadamard basis, displayed on the SLM (wavefront shaping unit **111**).

(21) Referring now to FIG. 4, a description will be given of the modulation pattern. FIG. 4 schematically illustrates the SLM (wavefront shaping unit **111**) and a modulation pattern. A modulated pattern **113** is displayed on the SLM, and the modulated pattern **113** is irradiated with the light **150** emitted from the light source **101**. A white portion **114** in the modulation pattern **113** represents a non-modulated area where the phase is not modulated, and a gray portion **115** represents a modulated area where the phase is modulated. In the modulated area, the phase is modulated by an arbitrary number of steps with respect to the non-modulated area, and the target **190** is irradiated with the irradiation light **151** modulated by the phase. Again, in step **S11**, the control unit **131** measures the feedback signal. Steps **S11** and **S12** are repeated the number of steps to search for the phase that maximizes the feedback signal, and SLM is set (N=1). Steps **S11** and **S12** are executed for an arbitrary number N of modulation patterns. After step **S10** (iterative processing of steps **S11** and **S12**) is executed the arbitrary number N of modulation patterns, the processing unit **130** ends the WFS for the first area. While FIG. 4 uses a two-dimensional SLM and illustrates an example of the two-dimensional modulation pattern **113**, a one-dimensional SLM may be used and a one-dimensional modulation pattern may be used.

(22) After step **S10** ends, the flow proceeds to step **S20**. In step **S20**, the processing unit **130** causes the irradiating unit **110** to irradiate the target **190** with the irradiation light **151** having the wavefront shaped in step **S10**, performs the DLS, and acquires a first measurement signal (second signal). More specifically, the processing unit **130** uses the detector **121**, causes the irradiating unit **110** to continuously irradiate the target **190** with the irradiation light (first irradiation light) **151** having the wavefront shaped (optimized) in step **S10**, and measures changes in light intensity of the exit light **161** over time. That is, the processing unit **130** causes the irradiating unit **110** to irradiate the target **190** with the first irradiation light having the shaped wavefront and measures the second signal (calculates correlation function $g(\tau)$). The correlation function $g(\tau)$ is expressed as in Equation (1) below:

$$(23) \quad g(\tau) = \frac{\langle I(t)I(t+\tau) \rangle}{\langle I(t) \rangle^2} \quad (1)$$

where $I(t)$ is the light intensity at time t , τ is the delay time, and $\langle \rangle$ is averaging.

(24) A hardware correlator may be used to calculate the correlation function $g(\tau)$. That is, the detecting unit **120** has a correlator, outputs the correlation function $g(\tau)$ from the output signal from the detector, and passes it to the data acquiring unit **132**. Alternatively, the correlation function $g(\tau)$ may be calculated using software. For example, the light intensity $I(t)$ received by detecting unit **120** may be received by data acquiring unit **132** and stored in the memory **134**. Next, the signal processing unit **133** may read the light intensity $I(t)$ from the memory **134** and calculate the

correlation function $g(\tau)$ by digital signal processing. FIG. 5 illustrates an example of the measured correlation function $g(\tau)$, that is, the measurement signal. In FIG. 5, a horizontal axis represents the delay time τ [s] and a vertical axis represents the correlation function $g(\tau)$. In this embodiment, a correlation function **141** measured in step **S20** is the first measurement signal (the second signal relating to the first area).

(25) Next, in steps **S30** and **S40**, the processing unit **130** performs the same processing as steps **S10** and **S20** for the detected position of the exit light **162**, respectively. More specifically, first, in step **S30**, the processing unit **130** sets the light intensity of the exit light **162** received by the detector **122** to a feedback signal and performs the WFS so as to maximize the feedback signal. As described above, the processing unit **130** measures the feedback signal using the detecting unit **120** in step **S31**, and shapes the wavefront of the irradiation light **151** using the wavefront shaping unit **111** in step **S32**. That is, the processing unit **130** feeds back the first signal (second feedback signal) from the detecting unit **120** that has received the exit light **162** that has propagated through a vascular area (second area) and exited from the target **190**, and causes the irradiating unit **110** to shape the wavefront of the irradiation light **151**. The control unit **131** repeats the iterative processing of steps **S31** and **S32** N times, which is the number of patterns, and optimizes the wavefront of the irradiation light **151** so as to maximize the intensity of the exit light **162** that has propagated through the second area (vascular area).

(26) Here, a constraint condition may be imposed so that the exit light **162** selectively propagates through the second area. For example, in optimizing the wavefront of the irradiation light **151**, the WFS of step **S32** may have a constraint condition that minimize the light intensity of the exit light **161** output from the detector **121** and maximizes the light intensity of the exit light **162** output from the detector **122**. The wavefront of the irradiation light **151** may be optimized under the constraint condition such that the signal intensity of the light that propagates through the vascular region **192** in the deep part of the target **190** and exits is maximized while the light that propagates through the surface layer of the target **190** is suppressed. Due to this WFS, the S/N ratio of the feedback signal to be measured can be improved while the propagation area is specified.

(27) After step **S30** ends, the flow proceeds to step **S40**. In step **S40**, the processing unit **130** causes the irradiating unit **110** to irradiate the target **190** with the irradiation light **151** having the wavefront shaped in step **S30**, performs the DLS, and acquires the correlation function $g(\tau)$ similarly to step **S20**. That is, the processing unit **130** causes the irradiating unit **110** to irradiate the target **190** with irradiation light (second irradiation light) **151** having the shaped wavefront and measures the second signal. For example, in step **S40**, a correlation function **142** illustrated in FIG. 5 is measured. This second signal is a second measurement signal (second signal relating to the second area). The measurement is thus completed, and all the measured data are stored in the memory **134** by the data acquiring unit **132**.

(28) Next, in step **S50**, the processing unit **130** compares and evaluates the first measurement signal measured in step **S20** and the second measurement signal measured in step **S40**. That is, the processing unit **130** acquires information on the target **190** by comparing a second signal based on the first irradiation light and a second signal based on the second irradiation light. The first measurement signal is the light that has propagated through the surface layer of the target **190** as described above, and since there are almost no elements that dynamically change scattering, the change in scattering over time is small. On the other hand, the second measurement signal is influenced by the vascular region **192** that dynamically changes scattering, and thus the change in scattering over time is large.

(29) Therefore, as illustrated in FIG. 5, it takes a long time τ for the correlation function of the first measurement signal to decay, whereas it takes a relatively short time for the correlation function of the second measurement signal to decay. Therefore, the first measurement signal whose results are relatively stable is set to reference data, and how much the second measurement signal is changed relative to the reference data can be relatively measured. For example, as a method of comparing

and evaluating these two signals, each correlation function $g(\tau)$ may be fitted with an exponential function $\exp(-\Gamma\tau)$ to calculate the decay coefficient Γ and the fitted correlation functions may be compared. The decay coefficients of the first measurement signal and the second measurement signal are set to $\Gamma_{\text{sub.1}}$ and $\Gamma_{\text{sub.2}}$, respectively, and changes of a ratio $\gamma = \Gamma_{\text{sub.2}}/\Gamma_{\text{sub.1}}$ or a difference $\gamma = \Gamma_{\text{sub.2}} - \Gamma_{\text{sub.1}}$ over time may be monitored. Although FIG. 5 illustrates the normalized correlation function, normalization is not necessary, and the measured correlation function can be evaluated through various analyzes other than the decay coefficient. Delay times until each correlation function decays to a certain value may be compared without fitting.

(30) The measurement flow according to this embodiment has thus been described. Pre-measurement for determining measurement conditions and measurement parameters may be performed before step S10. The detection positions p for the exit lights 161 and 162, WFS-related parameters, DLS measurement conditions, and the like may be determined based on the result of the pre-measurement.

(31) Here, this embodiment uses the DLS as a method for acquiring the information on the target 190 (biological physiological parameters). In the DLS, as described above, temporal variations in scattered light are reflected in the correlation function. For example, an element that provides a dynamic variation to the scattered light such as the bloodstream, and an element that provides a light intensity variation as an average bias component can be distinguished by delay time τ or frequency analysis. The latter is, for example, a system error on the light source or detection side, or an error component such as light absorption that does not locally change with time in the target 190. Therefore, obtaining a measurement signal using the DLS can reduce the influence of error factors and acquire biological information more robustly than simply measuring light intensity. The correlation function varies depending on physiological parameters of a living body, such as a scattering coefficient in blood and the blood flow (bloodstream) rate.

(32) Therefore, setting the vascular region 192 as a target to be measured and monitoring changes over time of the decay coefficient $\Gamma_{\text{sub.2}}$ of the second measurement signal can monitor changes in the blood scattering characteristic. For example, the more significant the blood scattering is, the larger the decay coefficient $\Gamma_{\text{sub.2}}$ is. Thus, the measurement flow illustrated in FIG. 3 using the measuring apparatus 100 according to this embodiment can monitor changes in the optical characteristic such as blood scattering. In addition, this method can measure a blood hemoglobin amount and other physiological parameters relating to blood such as oxygen saturation based on the DLS measurement result, by changing the wavelength of the light source 101 and.

(33) This embodiment can measure scattered light that reflects the optical characteristic of a specific area (first area or second area) inside the target 190 with an improved S/N ratio by applying the WFS and maximizing the feedback signal. Moreover, in measuring the physiological parameters of the target 190, this embodiment measures another signal, which is a DLS-based correlation function, instead of directly using the feedback signal. The feedback signal varies depending on the WFS measurement condition, such as the number of patterns N . In other words, since the feedback signal is a signal controllable on the device side, the measurement result cannot be directly used as a signal for measuring biological information. Therefore, this embodiment acquires another measurement signal that can utilize the effect of improving the feedback signal by the WFS.

(34) In a case where the effect of improving the S/N ratio by the WFS is lost in the measurements of steps S20 and S40, the WFS in steps S10 and S30 may be re-executed, and then the measurements in steps S20 and S40 may be re-executed. For example, the feedback signal may be monitored in step S20 or step S40 and the feedback signal may be compared with a preset threshold to determine whether to re-execute the WFS. The feedback signal can use another signal, as will be described in another embodiment below.

(35) In this embodiment, the processing unit 130 may measure the measurement signal (second signal) based on a parameter relating to the decay or delay time of the correlation function. The

first detector (detector **121**) may receive the exit light that has propagated through the first area (non-vascular area) as the specific area and exited from the target, and the second detector (detector **122**) may receive the exit light that has propagated through the second area (vascular area) as the specific area and exited from the target.

(36) In this embodiment, in a case where the signal intensity of the exit light **161** has a sufficient S/N ratio in step **S11**, step **S10** may be omitted. In this case, the measurement can be started with step **S20**. Although the two detectors **121** and **122** are provided in this embodiment, the number of detectors is not limited to two. In a case where a single detector is provided, steps **S10** and **S20** may be omitted, and only the second measurement signal may be measured by performing steps **S30** and subsequent steps. In this case, step **S50** does not perform the comparison or evaluation, and can output the result of the second measurement signal. Three or more detectors may be provided so that a measurement area of the target **190** is limited for each detector, the WFS may be executed for each detector to acquire the measurement signal, and the results may be evaluated.

Second Embodiment

(37) A description will now be given of a second embodiment according to the disclosure. The measuring apparatus according to this embodiment has the same basic configuration as that of FIG. **1**, but the detecting unit **120** includes a single detector **121**. The measuring method according to this embodiment follows the flowchart illustrated in FIG. **3**. This embodiment uses the single detector **121** to perform two measurements for the first non-vascular area and the second vascular area described in the first embodiment, and compare and evaluate the two measurement results.

(38) First, in step **S10** of FIG. **3**, the control unit **131** executes the WFS so as to measure the first area near the surface layer of the target **190**. A signal based on the correlation function $g(\tau)$ obtained by executing the DLS is used as the feedback signal for the WFS. FIG. **6** explains a correlation function **143** obtained by measuring the time variation of the intensity of the exit light **161** with the detector **121**. In FIG. **6**, a horizontal axis represents the delay time τ , and a vertical axis represents the correlation function $g(\tau)$. Here, a part of the correlation function **143** may be used as the feedback signal. For example, the part of the correlation function **143** may be a value of the correlation function (feedback signal $g(\tau_{\text{sub.1}})$) when the delay time $T_{\text{sub.1}}$ has passed.

(39) As illustrated in the flowchart of FIG. **3**, this feedback signal $g(\tau_{\text{sub.1}})$ is monitored in step **S11**, the phase of the modulation pattern is changed in step **S12**, and the feedback signal is again monitored in step **S11**. For the fast iterative processing of step **S10**, the acquisition time of the feedback signal may be shortened. Therefore, the delay time $T_{\text{sub.1}}$ is made as short as possible. On the other hand, for the second area in which a dynamic change in the target **190** is expected, the setting may be made within a detectable range of the change in the feedback signal value $g(\tau_{\text{sub.1}})$.

(40) In step **S12**, the control unit **131** monitors the feedback signal $g(\tau_{\text{sub.1}})$ and causes the irradiating unit **110** to shape the wavefront of the irradiation light **151** so that the feedback signal $g(\tau_{\text{sub.1}})$ does not increase or attenuate. In other words, the wavefront is optimized in step **S10** so that the detecting unit **120** can detect the propagation light that has avoided a dynamically changing region that affects the correlation function (such as the vascular area in the second area) as fully as possible. In other words, the wavefront is optimized so that the exit light that mainly propagates through the non-vascular area of the first area is measured.

(41) Next, in step **S20**, the control unit **131** causes the irradiating unit **110** to irradiate the target **190** with the irradiation light **151** having the wavefront shaped in step **S10** and measures as a measurement signal a light intensity $V_{\text{sub.1}}$ (first measurement signal) different from the feedback signal $g(\tau_{\text{sub.1}})$ using the detector **121**.

(42) Next, in step **S30**, similarly to step **S20**, the control unit **131** monitors the feedback signal $g(\tau_{\text{sub.1}})$ and executes the WFS. More specifically, the control unit **131** acquires the feedback signal in step **S31**, and shapes the wavefront of the irradiation light **151** so that the feedback signal becomes smaller in step **S32**. That is, in step **S30**, the control unit **131** performs the WFS so that the

feedback signal $g(\tau_{\text{sub.1}})$ becomes smaller (attenuates). Therefore, the wavefront is optimized so that the detecting unit **120** detects propagation light that has been affected by the dynamic scattering changes due to blood flow or the like. In other words, the WFS is executed using the vascular area of the second area to be measured.

(43) Next, in step **S40**, the control unit **131** causes the irradiating unit **110** to irradiate the target **190** with the irradiation light **151** having the wavefront shaped in step **S30** and measures the light intensity $V_{\text{sub.2}}$ (second measurement signal) as the measurement signal. In the measurement of step **S40**, the control unit **131** may execute the WFS of step **S30** as needed to update the wavefront of the irradiation light **151** as needed, and execute the measurement of step **S40**.

(44) Next, in step **S50**, the control unit **131** compares and evaluates the two measurement signals obtained in steps **S20** and **S40**. As described in the first embodiment, the ratio or difference between the light intensities $V_{\text{sub.1}}$ and $V_{\text{sub.2}}$ may be evaluated. Alternatively, the measurement result may be analyzed based on a model such as a transport equation and a diffusion equation, and the result may be evaluated. For example, in a case where an absorption wavelength of a specific target such as blood hemoglobin is measured with the irradiation light **151**, the absorbance or the relative concentration of the target may be evaluated based on the measurement signal.

(45) Similarly to the first embodiment, this embodiment can perform pre-measurement before step **S10** to determine a parameter such as a measurement condition in advance. In this embodiment, the detection position (distance ρ) illustrated in FIG. 2 may be fixed or variable. In the latter case, the detection positions measured in steps **S10** and **S20** and steps **S30** and **S40** may be changed based on the results of pre-measurement.

(46) As a variation of this example, the autofluorescence of the target **190** may be used. A wavelength range of 300 to 400 nm is used for the light source **101**. For example, in steps **S10** and **S30** of FIG. 3, the detector **121** receives light having a wavelength that is excited by the irradiation light **151** and emitted by autofluorescence as a WFS feedback signal. The wavefront shaping unit **111** can shape the wavefront of the irradiation light **151** so that the fluorescence intensity received by the detector **121** increases. In the subsequent measurements in steps **S20** and **S40**, the correlation function may be evaluated by measuring variations with time in the light intensity of autofluorescence based on the DLS. Here, in steps **S10** and **S30**, the detecting unit **120** may have an excitation light cut filter such that only fluorescence is measured by the detector **121**. Depending on the type of autofluorescence, the depth of the light source inside the target **190** can be estimated. For example, advanced glycation end products, NADH, and the like may be used as autofluorescence. The light source **101** may be a light source having a plurality of wavelengths in order to handle a plurality of phosphors and various measurement items.

(47) In the measurements of steps **S20** and **S40**, the detector **121** may receive the light emitted at the wavelength of the excitation light, the variations with time may be measured based on the DLS, and the correlation function may be evaluated.

(48) Alternatively, the feedback signal for the WFS is not limited to the above autofluorescence, and a fluorescent contrast agent may be administered and its luminescence may be used. Steps **S20** and **S40** may directly measure the feedback signal as the second signal, and the light from the light source can be concentrated and irradiated onto a light emitting area inside the target **190**. The effect of this concentrative irradiation may be used to treat lesions inside the target **190**. In this case, a laser may be used as the light source.

(49) The measuring apparatus according to each embodiment may not have the reflection type configuration as illustrated in FIG. 1 and may have a transmission type configuration. Various combinations of the feedback signal (first signal) for executing the WFS and the measurement signal (second signal) for acquiring target information (physiological information on a living body) are conceivable and are not limited the above embodiments. In each embodiment, as the WFS effects, the improvement of the S/N ratio obtained as a result of scattering suppression and light concentration effect to a more localized area in the scattering medium are used to measure a living

body. At this time, since the feedback signal used for the WFS is operated and controlled on the measuring apparatus side, each embodiment acquires biometric information by measuring another measurement signal that the WFS effect reaches without directly using the above feedback signal. (50) As described above, the measuring apparatus **100** according to each embodiment is a measuring apparatus configured to acquire information on a target, and includes an irradiating unit **110**, a detecting unit **120**, and a processing unit **130**. The processing unit **130** causes the irradiating unit **110** to shape the wavefront of the irradiation light by feeding back the first signal (feedback signal) output from the detecting unit **120** that has received the exit light. The processing unit **130** acquires information about the specific area using the second signal (measurement signal) output from the detecting unit **120** that has received the exit light from the target by irradiating the target with the irradiation light having the shaped wavefront.

(51) The first signal or the second signal may be a light intensity measured based on a distance between the irradiating position where the irradiating unit **110** irradiates the target with the irradiation light and the detection position where the detecting unit **120** detects the exit light. Alternatively, the first signal or second signal may be a signal based on a correlation function calculated by dynamic light scattering (DLS), or a fluorescence intensity emitted by autofluorescence (or fluorescent dye). The processing unit **130** may monitor changes in the information on the target by comparing the second signal measured for a first area and the second signal measured for a second area that are at least two different specific areas.

(52) The irradiating unit **110** may shape the wavefront of the irradiation light so as to reduce the first signal obtained for the first area, and shapes the wavefront of the irradiation light so as to increase the first signal obtained for the second area. The target information may be physiological information inside a living body, the first area may be a non-vascular area in the living body, and the second area may be a vascular area in the living body. The processing unit **130** may use the second signal for the non-vascular area as reference data, and monitors a change in the second signal for the vascular area relative to the reference data.

Other Embodiments

(53) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read-only memory (ROM), a storage of distributed computing systems, an optical disc (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(54) Each embodiment can measure a more local physiological parameter in a living body with an excellent S/N ratio by suppressing scattering and limiting a measurement area in the living body as a target. Therefore, each embodiment can provide a measuring apparatus, a measuring method, and a storage medium, each of which can measure target information with high accuracy using light.

(55) While the disclosure has been described with reference to embodiments, it is to be understood

that the disclosure is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(56) For example, each embodiment has described a measuring apparatus and a measuring method configured to acquire physiological information in a living body as information on a target, but the disclosure is not limited to this example, and can be applied to other target measurements.

Claims

1. A measuring apparatus configured to acquire information on a target, the measuring apparatus comprising: an irradiating unit configured to irradiate a specific area of the target with irradiation light; a detecting unit configured to receive exit light from the target which is caused by irradiating the specific area with the irradiation light; and a processing unit configured to process a signal output from the detecting unit, wherein the processing unit causes the irradiating unit to shape a wavefront of the irradiation light by feeding back a first signal output from the detecting unit that has received the exit light, and wherein the processing unit acquires information about the specific area using a second signal output from the detecting unit that has received exit light from the target which is caused by irradiating the specific area with irradiation light having the shaped wavefront, wherein the processing unit monitors changes in the information on the target by comparing the second signal measured for a first area and the second signal measured for a second area, and the first area and the second area are two different specific areas, and wherein the processing unit uses the second signal for the first area as reference data, and monitors a change in the second signal for the second area relative to the reference data.
2. The measuring apparatus according to claim 1, wherein the first signal or the second signal is a light intensity measured based on a distance between an irradiating position where the irradiating unit irradiates the target with the irradiation light and a detection position where the detecting unit detects the exit light, a signal based on a correlation function calculated by dynamic light scattering, or a fluorescence intensity emitted by fluorescent dye.
3. The measuring apparatus according to claim 1, wherein the irradiating unit shapes the wavefront of the irradiation light so as to maximize or minimize the first signal.
4. The measuring apparatus according to claim 1, wherein the irradiating unit includes a spatial light modulator, and shapes the wavefront of the irradiation light through phase modulation or amplitude modulation using the spatial light modulator based on a base pattern.
5. The measuring apparatus according to claim 1, wherein the irradiating unit shapes the wavefront of the irradiation light so as to reduce the first signal obtained for the first area, and wherein the irradiating unit shapes the wavefront of the irradiation light so as to increase the first signal obtained for the second area.
6. The measuring apparatus according to claim 1, wherein the information on the target is physiological information inside a living body, wherein the first area is a non-vascular area in the living body, and wherein the second area is a vascular area in the living body.
7. The measuring apparatus according claim 1, wherein the detecting unit includes a first detector and a second detector, wherein the first detector receives the exit light that has propagated through the first area as the specific area and exited from the target, and wherein the second detector receives the exit light that has propagated through the second area as the specific area and exited from the target.
8. The measuring apparatus according to claim 1, wherein the processing unit causes the irradiating unit to shape the wavefront of the irradiation light by feeding back the first signal from the detecting unit that has received the exit light that has propagated through the first area as the specific area and exited from the target, wherein the processing unit measures the second signal by causing the irradiating unit to irradiate the target with first irradiation light that has the shaped

wavefront, wherein the processing unit causes the irradiating unit to shape the wavefront of the irradiation light by feeding back the first signal obtained from the detecting unit that has received exit light that has propagated through the second area as the specific area and exited from the target, wherein the processing unit measures the second signal by causing the irradiating unit to irradiate the target with second irradiation light that has the shaped wavefront, and wherein the processing unit acquires the information on the target by comparing the second signal based on the first irradiation light and the second signal based on the second irradiation light.

9. A measuring apparatus configured to acquire information on a target, the measuring apparatus comprising: an irradiating unit configured to irradiate a specific area of the target with irradiation light; a detecting unit configured to receive exit light from the target which is caused by irradiating the specific area with the irradiation light; and a processing unit configured to process a signal output from the detecting unit, wherein the processing unit causes the irradiating unit to shape a wavefront of the irradiation light by feeding back a first signal output from the detecting unit that has received the exit light, wherein the processing unit acquires information about the specific area using a second signal output from the detecting unit that has received exit light from the target which is caused by irradiating the specific area with irradiation light having the shaped wavefront, and wherein the processing unit measures the second signal based on a parameter relating to decay or delay time of a correlation function.

10. A measuring method configured to acquire information on a target, the measuring method comprising: irradiating a specific area of the target with irradiation light; receiving first exit light from the target, which is caused by irradiating the specific area with the irradiation light, and outputting a first signal based on the received first exit light; shaping a wavefront of the irradiation light by feeding back the first signal; and receiving second exit light from the target, which is caused by irradiating the specific area with the irradiation light having the shaped wavefront, outputting a second signal based on the received second exit light, and acquiring information about the specific area using the second signal, wherein changes in the information on the target are monitored by comparing the second signal measured for a first area and the second signal measured for a second area, and the first area and the second area are two different specific areas, and wherein the second signal for the first area is used as reference data, and a change in the second signal for the second area is monitored relative to the reference data.

11. A non-transitory computer-readable storage medium storing a program that causes a computer to execute the measuring method according to claim 10.
